	Scotch Student ID #			
Circle the relevant digits	0	0	0	0
	1	1	1	1
	2	2	2	2
	3	3	3	3
	4	4	4	4
	5	5	5	5
	6	6	6	6
	7	7	7	7
	8	8	8	8
	9	9	9	9

Teacher's Name

Scotch College

CHEMISTRY

SAC 1 – FUELS, GASES & THERMOCHEMISTRY

Date: Thursday, March 17th

Reading Time	0 minutes
Writing Time	45 minutes

Task Sections	Marks	Your Marks
Total Marks	40	

General Instructions

- Write your student number in the space provided above on this page.
- Please complete all questions in the spaces provided.

Allowed Materials

- A VCAA Chemistry Data Book has been provided.
- Students are permitted to bring into the SAC: pens, pencils, highlighters, erasers, sharpeners, rulers and one scientific calculator.

At the end of the task…

- Please ensure your student number (not name) has been filled in at the top of the page and that you have written your teacher's name.

Electronic Devices

Students are **not** allowed to have a mobile phone, smart watch and/or any other unauthorised electronic device in the SAC, unless it is TURNED OFF and is placed on the front teacher desk.

Q1. *(1 mark)*
What is the enthalpy change, **in J**, when 5 kg of water is heated from 10°C to 18°C?

A. $5 \times 4.18 \times 8$

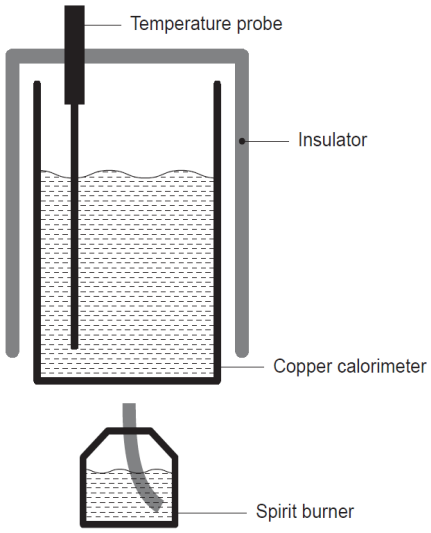
B. $5 \times 10^3 \times 4.18 \times 8$

C. $5 \times 4.18 \times (273 + 8)$

D. $5 \times 10^3 \times 4.18 \times (273 + 8)$

Q2. (*1 mark)*

The enthalpy of combustion of a fuel was determined using the calorimeter shown.



The final result was **lower** than the literature value. Which factors could have contributed to this error?

I. Not all heat from the combustion was transferred to the calorimeter.
II. Incomplete combustion occurred.
III. The temperature probe touched the bottom of the calorimeter.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

Q3. (*1 mark)*

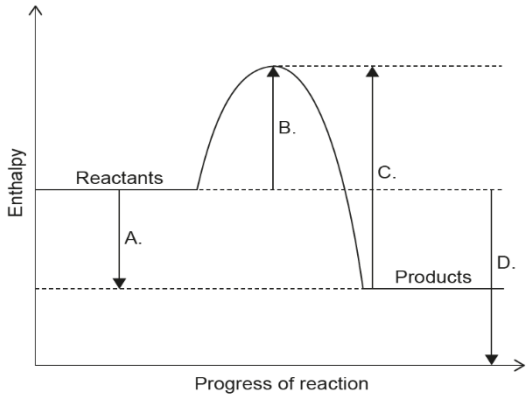
On the energy profile shown below, which letter (A-D) represents the enthalpy change of reaction, Δ*H*?

A.

B.

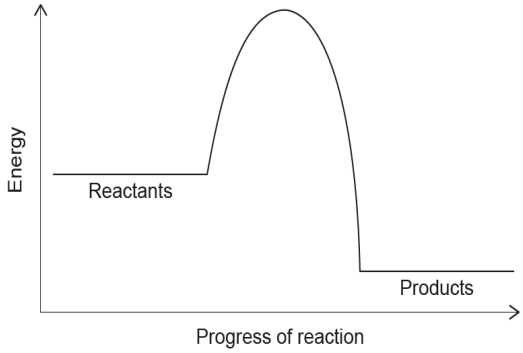
C.

D.



Q4. *(1 mark)*

The potential energy profile of a reaction is shown.

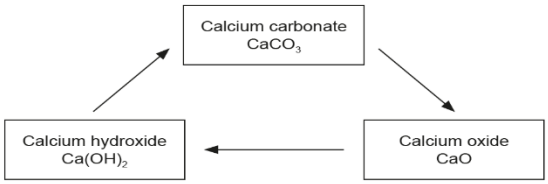


What can be determined about stability and energy change from the potential energy profile shown?

	More Stable	**Reaction**
A.	Reactants	Exothermic
B.	Reactants	Endothermic
C.	Products	Exothermic
D.	Products	Endothermic

Q5. *(8 marks)*

Limestone can be converted into a variety of useful commercial products through the lime cycle. Limestone contains high percentages of calcium carbonate, $CaCO_3$, $M(CaCO_3) = 100.1$ g mol^{-1}.



a. Calcium carbonate is heated to produce calcium oxide, CaO. (3 marks)

$$CaCO_3\,(s) \rightarrow CaO\,(s) + CO_2\,(g) \quad \Delta H = 177.8 \text{ kJ mol}^{-1}$$

Calculate the volume of carbon dioxide produced at 150 kPa and 100 °C when 555 g of calcium carbonate decomposes.

b. Calcium hydroxide reacts with carbon dioxide to reform calcium carbonate. (3 marks)

$$Ca(OH)_2\,(aq) + CO_2\,(g) \rightarrow CaCO_3\,(s) + H_2O\,(l) \quad \Delta H = -69.1 \text{ kJ mol}^{-1}$$

Determine the mass, in g, of $CaCO_3$ (s) produced by reacting 2.41 L of 2.33×10^{-2} M of $Ca(OH)_2$ (aq) with 0.750 L of CO_2 (g) at SLC.

c. Given the following thermochemical equations, determine the thermochemical equation for the formation of calcium hydroxide ($Ca(OH)_2$) from calcium oxide (CaO) and water. (2 marks)

$$Ca(OH)_2\,(aq) + CO_2\,(g) \rightarrow CaCO_3\,(s) + H_2O\,(l) \qquad \Delta H = -69.1 \text{ kJ mol}^{-1}$$

$$CaCO_3\,(s) \rightarrow CaO\,(s) + CO_2\,(g) \qquad \Delta H = +177.8 \text{ kJ mol}^{-1}$$

Q6. *(12 marks)*

a. Ethanol is a biofuel that can be mixed with petrol.

Write the **thermochemical** equation for the complete combustion of liquid ethanol. (2 marks)

b. All fuels produce carbon dioxide during combustion. Explain, including a suitable equation, why bioethanol can be considered to be more carbon neutral than ethanol derived from fossil fuels. (3 marks)

c. Biodiesel can be made when ethanol reacts with vegetable oil. What other product is formed in this process? (1 mark)

Methyl palmitate, $C_{17}H_{34}O_2$, is a component of a type of biodiesel. It is liquid at room temperature. The molar enthalpy of combustion was determined using a bomb calorimeter. The calorimeter was calibrated by passing 3.40 amps and 5.61 volts through an electric heater for 240 seconds. The temperature of the calorimeter rose by 1.75 ^{o}C.

d. Calculate the calibration factor of the calorimeter. (3 marks)

A 0.578 g sample of the methyl palmitate was then burned in excess oxygen in the calorimeter and the temperature rose by 6.19 ^{o}C. $M(C_{17}H_{34}O_2) = 270.0$ g mol^{-1}.

e. Calculate the amount of energy, in kJ, absorbed by the calorimeter when the sample of methyl palmitate was burned. (1 mark)

f. Calculate the amount of energy released, in kJ, by the combustion of 1.00 mol of methyl palmitate. (2 marks)

Q7. *(12 marks)*

Vegetable oils require conversion to biodiesel for use in internal combustion engines.

a. A student obtained a sample of petrodiesel and biodiesel for the purpose of investigating their properties. Unfortunately, when they placed the samples into iced water, the labels came off. Use the results below to identify the samples. Fill in the name of tube A below. (1 mark)

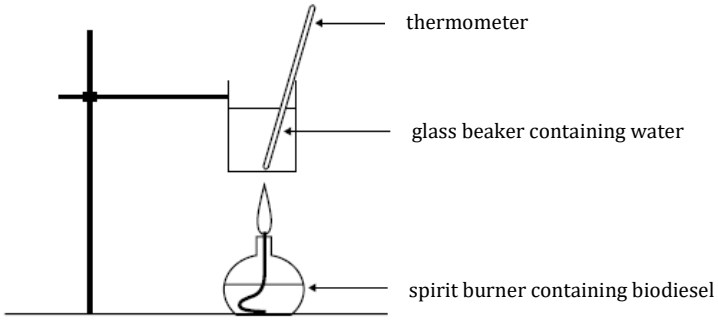
	Flow rate (L min^{-1})	
Temperature (°C)	Tube A	Tube B
0	Crystalised	0.11
Room temp	0.16	0.25
30	0.37	0.55

Tube A _____

b. What structural difference between petrodiesel and biodiesel are most likely to be responsible for the property named in 7a. Explain your answer by describing the structure of both fuels.

(3 marks)

Having tried to determine the energy content of biodiesel by burning it in a crucible sitting in water and being disappointed by the amount of heat lost, the student poured the remaining biodiesel into a spirit burner and tried again.



The results are shown below.

Mass of water (g)	198.203
Mass of biodiesel burnt (g)	2.076
Initial temperature of water (°C)	19.2
Final temperature of water (°C)	25.8

c. Calculate the energy, in Joules, absorbed by the water. (2 marks)

d. Determine the heat of combustion of the biodiesel in kJ g^{-1}. (1 mark)

e. The results were still lower than the value listed in the text book, so the student decided to make two changes to the set up shown above. What are two changes that would **reduce energy loss** in this apparatus? (2 marks)

f. While packing the equipment away the student noticed that the bottom of the glass beaker had turned black (covered in carbon soot). Explain this observation by writing a balanced chemical equation using the fuel octane, C_8H_{18}. States are not required. (1 mark)

Q8. *(4 marks)*

Pig and cattle farmers across Australia are using the biogas from their animal's waste to produce the electricity for their farms. An average pig will produce 2.98 kg of methane gas per day.

a. Calculate the maximum energy, in kJ, that could be generated by combustion of the methane produced by an average pig in one year (365 days). (1 mark)

b. The methane gas is collected in tanks before being used in a combustion engine to produce electricity. A farmer setting up a biogas collection system of flexible polymer tanks. What volume, in ML, would they expect to collect from 200 pigs, ***per day***, if the gas is stored at SLC? (2 marks)

c. While considerably more expensive, some farmers store their methane by increasing the pressure to 4550 kPa. If they keep the temperature at 25.0 ^{o}C, what volume, in ML, would the volume from question **8b** (above) of methane occupy? (1 mark)

End of SAC